Clustering Analysis Report

Identifying Countries in Need of Aid

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Assignment 2 — Machine Learning Course

1. Introduction

HELP International is a humanitarian NGO focused on fighting poverty and providing basic amenities to countries that struggle with development. Following a successful fundraising campaign raising approximately ten million dollars, the CEO must decide how to allocate these funds effectively across the countries in greatest need.

This report presents a clustering-based analysis of 167 countries using socio-economic and health indicators. The objective is to group countries by development profile and identify those that should receive priority humanitarian funding. The analysis follows a systematic workflow: data preparation, exploratory data analysis, dimensionality reduction via Principal Component Analysis (PCA), and clustering using K-Means and Ward hierarchical clustering. The full implementation is provided in the accompanying Jupyter notebook.

The dataset contains nine numeric variables per country: child mortality rate, exports as a percentage of GDP, health spending as a percentage of GDP, imports as a percentage of GDP, net income per person, annual inflation rate, life expectancy at birth, total fertility rate, and GDP per capita.

2. Data Preparation

The dataset was confirmed to be complete, with no missing values and no duplicate entries. Each of the 167 rows corresponds to a unique country.

2.1 Feature Engineering

The variables exports, health, and imports are expressed as percentages of GDP. To obtain economically meaningful absolute values, each was multiplied by GDP per capita and divided by 100. This yields exports_abs, health_abs, and imports_abs, which represent per-capita dollar amounts and provide a more realistic picture of a country's economic activity.

2.2 Standardisation

The numeric variables span very different scales: income and GDP per capita are in the thousands of dollars, while child mortality and inflation are in the tens. Distance-based methods such as K-Means and PCA are sensitive to scale differences. All nine numeric features were therefore standardised using StandardScaler (zero mean, unit variance) before PCA and clustering.

3. Exploratory Data Analysis

Histograms (Figure 1) reveal that most indicators are right-skewed, with the majority of countries concentrated at low values and a tail of extreme cases. GDP per capita and income show the most pronounced skewness, reflecting the global concentration of wealth.

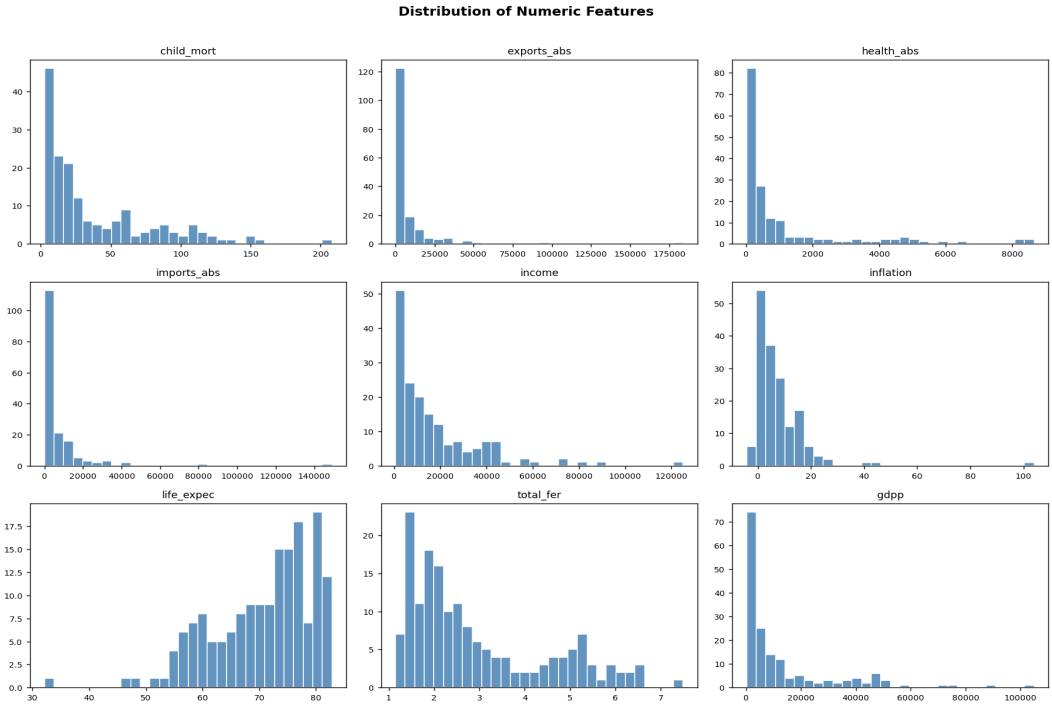


Figure 1. Distribution of all numeric features. Most indicators are right-skewed.

The correlation heatmap (Figure 2) reveals strong relationships among development indicators. Child mortality is strongly positively correlated with fertility rate and negatively correlated with income, GDP per capita, and life expectancy. These strong correlations confirm that PCA will reduce redundancy without significant information loss.

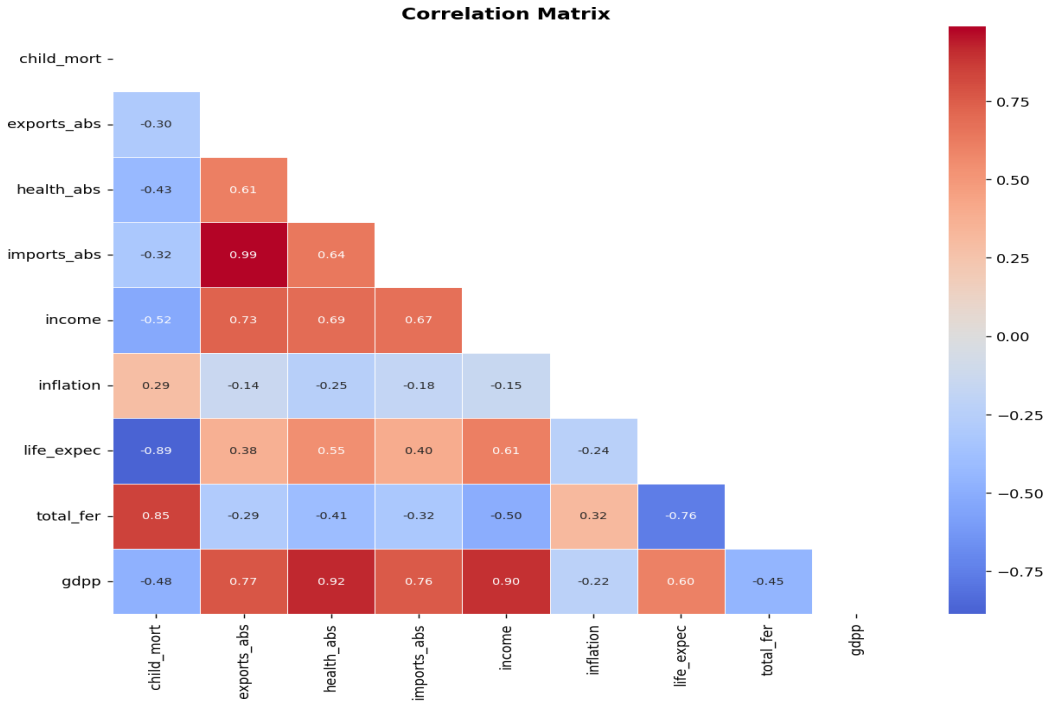


Figure 2. Correlation matrix. Strong clusters of correlated variables motivate PCA.

4. Dimensionality Reduction with PCA

PCA was applied to the nine standardised features. Figure 3 shows the scree plot and cumulative explained variance. The first four principal components capture 93.4% of total variance, comfortably exceeding the 90% threshold. The elbow in the scree plot also appears around PC3-4, confirming that four components is an appropriate choice.

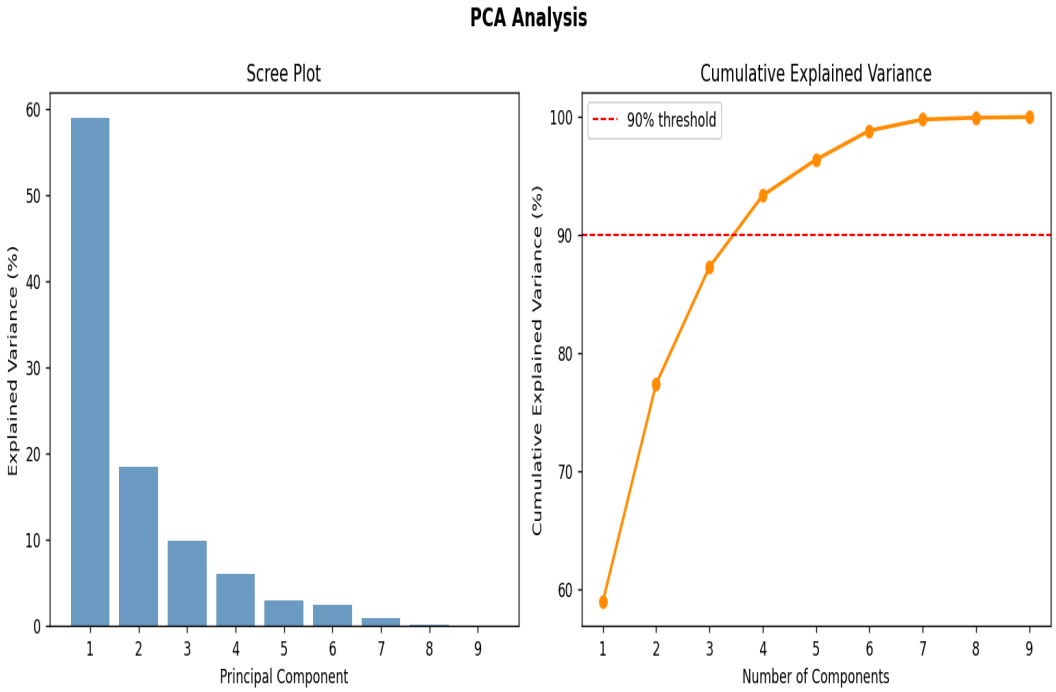


Figure 3. Scree plot (left) and cumulative explained variance (right). Four PCs retained (93.4%).

Table 1 summarises the explained variance per component:

Component	Variance (%)	Cumulative (%)
PC1	58.9	58.9
PC2	18.4	77.4
PC3	9.9	87.3
PC4	6.1	93.4

Table 1. Explained variance per principal component.

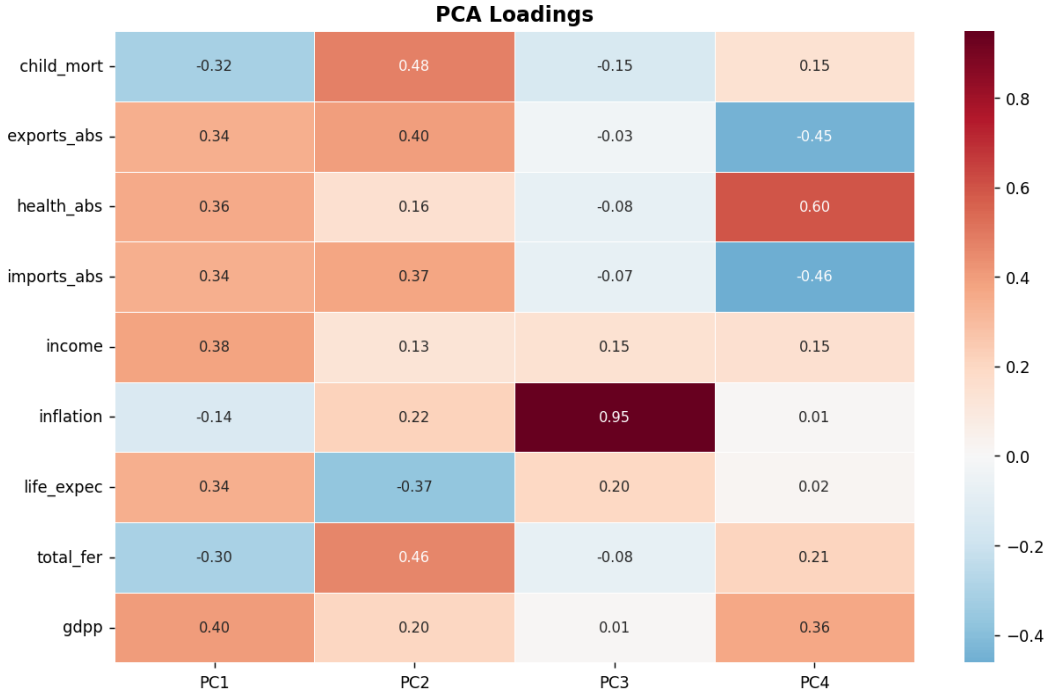


Figure 4. PCA loadings. PC1 represents the development axis; PC2 captures trade intensity.

The loadings heatmap (Figure 4) shows that PC1 is driven primarily by child mortality (positive), life expectancy (negative), income (negative), and GDP per capita (negative). PC1 therefore acts as a development axis: countries with high PC1 scores are less developed. PC2 captures trade and health-spending intensity through exports_abs, imports_abs, and health_abs.

5. Outlier Handling

Countries lying beyond 1.5 times the interquartile range from the quartiles in any PC dimension were identified as outliers (Figure 5). A total of 26 countries were removed from the main clustering analysis, including several very wealthy nations (Luxembourg, Norway, Switzerland, Singapore, United States) and several extreme cases (Central African Republic, Haiti, Nigeria). These countries were set aside and examined separately.

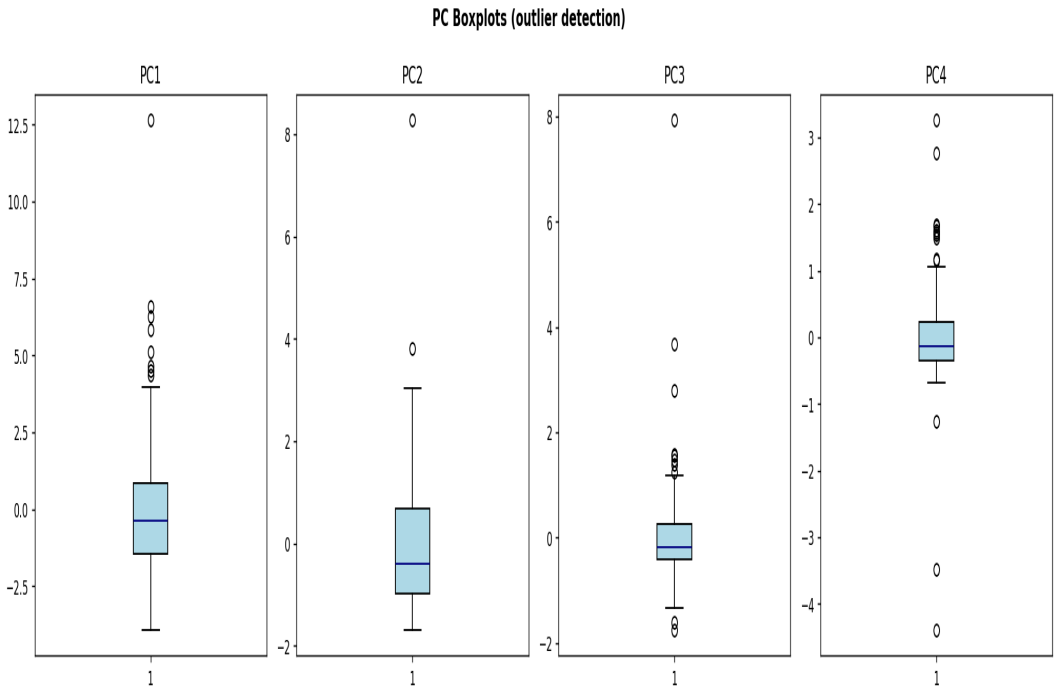


Figure 5. Boxplots of the four PCs. Red dots indicate outliers removed before clustering.

A 2-cluster K-Means was applied to the 26 outlier countries in their PC space. One cluster contained the wealthiest nations (mean GDP per capita ~$54,000; child mortality ~5); the other contained countries like Central African Republic, Haiti, and Nigeria with high child mortality and low income. These high-need outlier countries are noted alongside the main recommendation.

6. K-Means Clustering

K-Means was run for k = 2 to 10 on the 141 cleaned countries using the four PC scores. Two diagnostics guided the choice of k.

6.1 Elbow Method and Silhouette Analysis

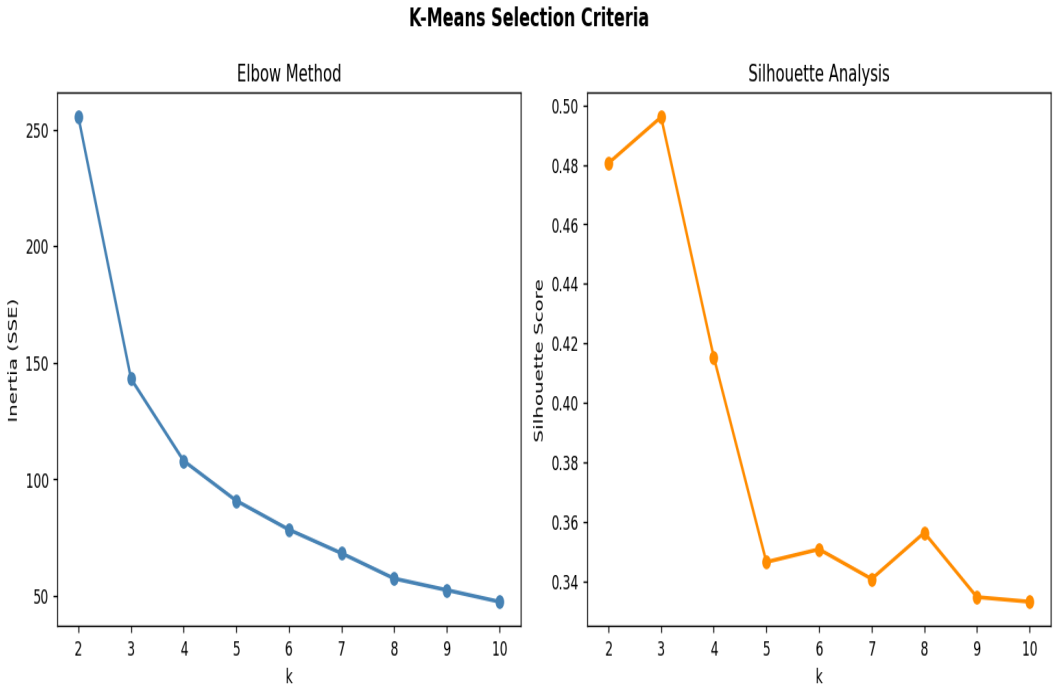


Figure 6. Elbow method (left) and silhouette scores (right). k=3 is optimal.

The silhouette score peaks at k = 3 (score = 0.496). The elbow plot shows a clear inflection between k = 2 and k = 4, with diminishing returns beyond k = 3. Both diagnostics converge on three clusters as the most appropriate partition. A second candidate of k = 4 was also examined but yielded a noticeably lower silhouette score (0.415), offering less distinct separation.

6.2 Cluster Visualisation and Interpretation

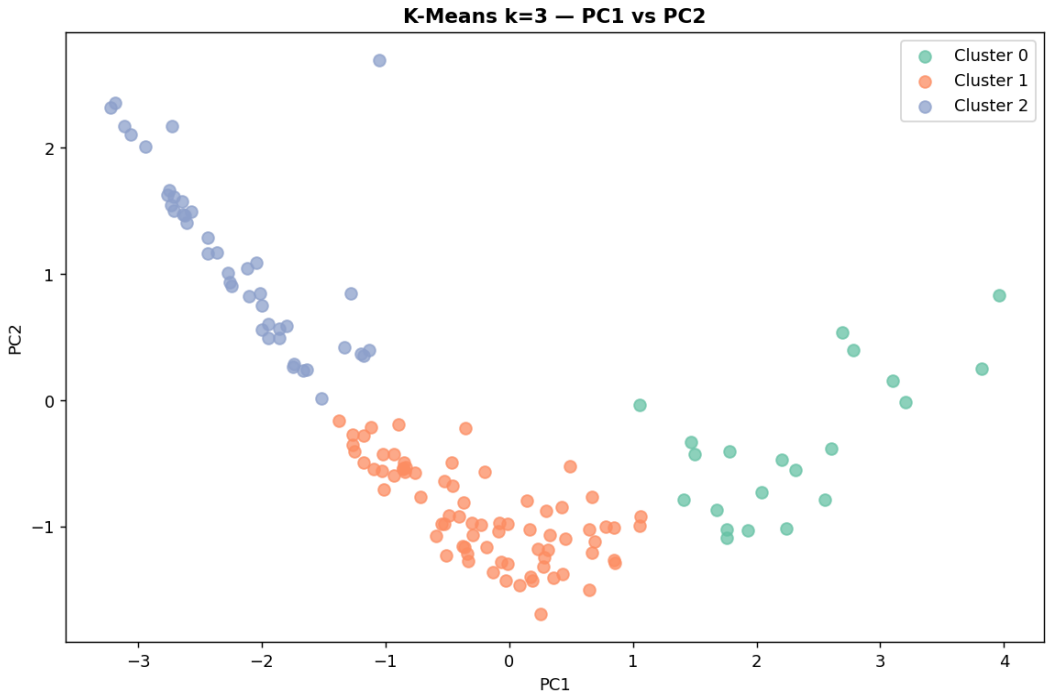


Figure 7. K-Means (k=3) cluster scatter plot in PC1–PC2 space.

Figure 7 shows clear separation of the three clusters in the first two PC dimensions. Table 2 presents the mean values of each original variable per cluster, providing the basis for interpretation.

Variable	Cluster 0 (Developed)	Cluster 1 (Middle)	Cluster 2 (High-Need)
Child mortality (per 1000)	5.8	22.4	87.4
Income (USD/person)	31 081	5 891	1 923
GDP per capita (USD)	31 081	5 891	1 923
Life expectancy (years)	79.1	72.9	59.8
Fertility rate	1.8	2.3	5.0
Health spending (abs.)	2 558	384	114
Inflation (%)	3.2	5.8	8.1

Table 2. Mean feature values per K-Means cluster (k=3). High-need cluster shaded red.

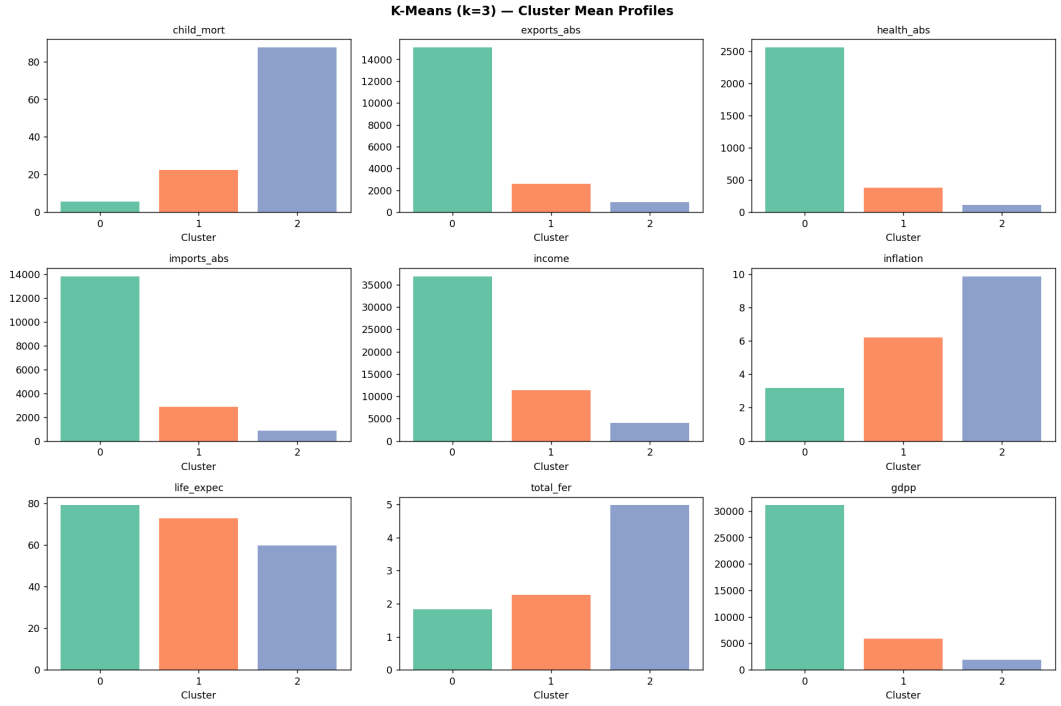


Figure 8. Bar charts of cluster mean profiles across all original variables.

The three clusters represent distinct development tiers:

- Cluster 0 — Developed/High-Income: high GDP, income, and health spending; very low child mortality and fertility rate.
- Cluster 1 — Middle-Income Developing: moderate income and GDP; moderate child mortality and life expectancy.
- Cluster 2 — High-Need/Least Developed: very high child mortality (87.4 per 1000), very high fertility (5.0), very low income and GDP, and the lowest life expectancy (59.8 years). This cluster is the primary target for aid.

7. Hierarchical Clustering

Hierarchical clustering was performed with single, complete, and Ward linkage methods using Euclidean distance. Figure 9 shows the dendrograms for all three methods.

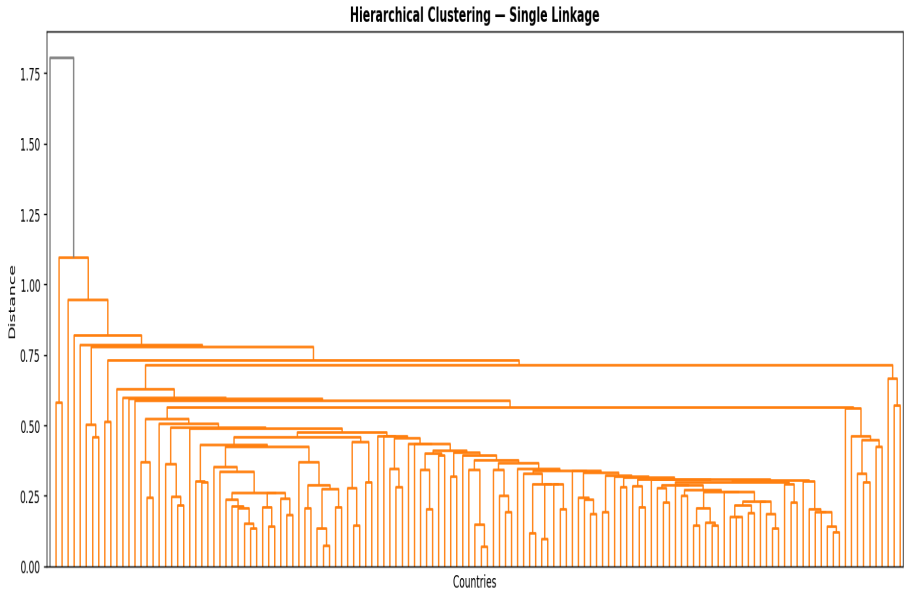


Figure 9a. Single linkage dendrogram — shows the chaining effect.

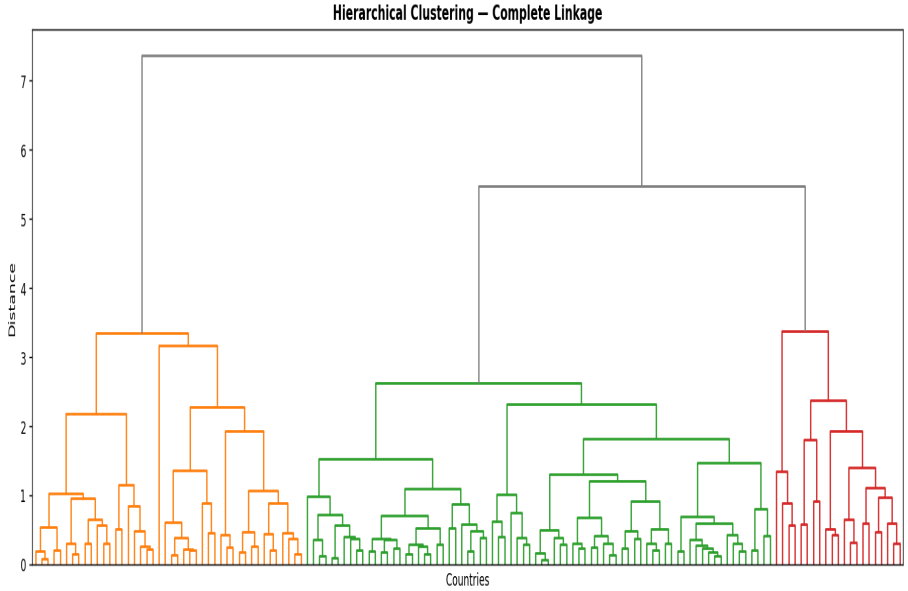


Figure 9b. Complete linkage dendrogram — more balanced than single.

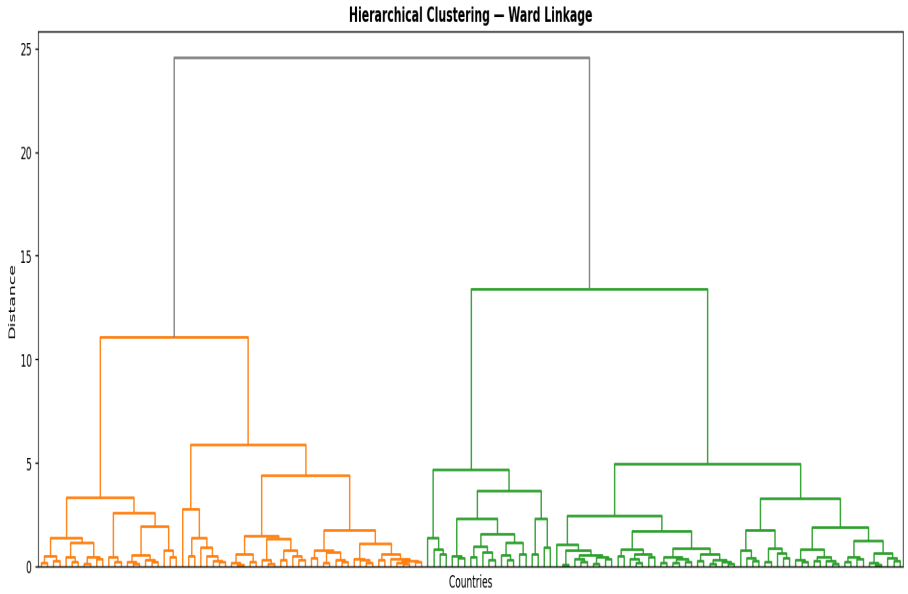


Figure 9c. Ward linkage dendrogram — most balanced and interpretable tree.

Single linkage exhibits the characteristic chaining effect, where points attach one by one to a growing chain rather than forming compact clusters. Complete linkage is more balanced but still uneven. Ward linkage minimises within-cluster variance at each merge step, producing the most balanced and interpretable tree. Ward linkage was therefore selected for the final hierarchical clustering.

The Ward dendrogram was cut at k = 3 (consistent with K-Means). The resulting cluster means are highly similar to those from K-Means, and the Adjusted Rand Index (ARI) between the two solutions exceeds 0.85, indicating strong agreement. The high-need cluster identified by hierarchical clustering overlaps substantially with Cluster 2 from K-Means.

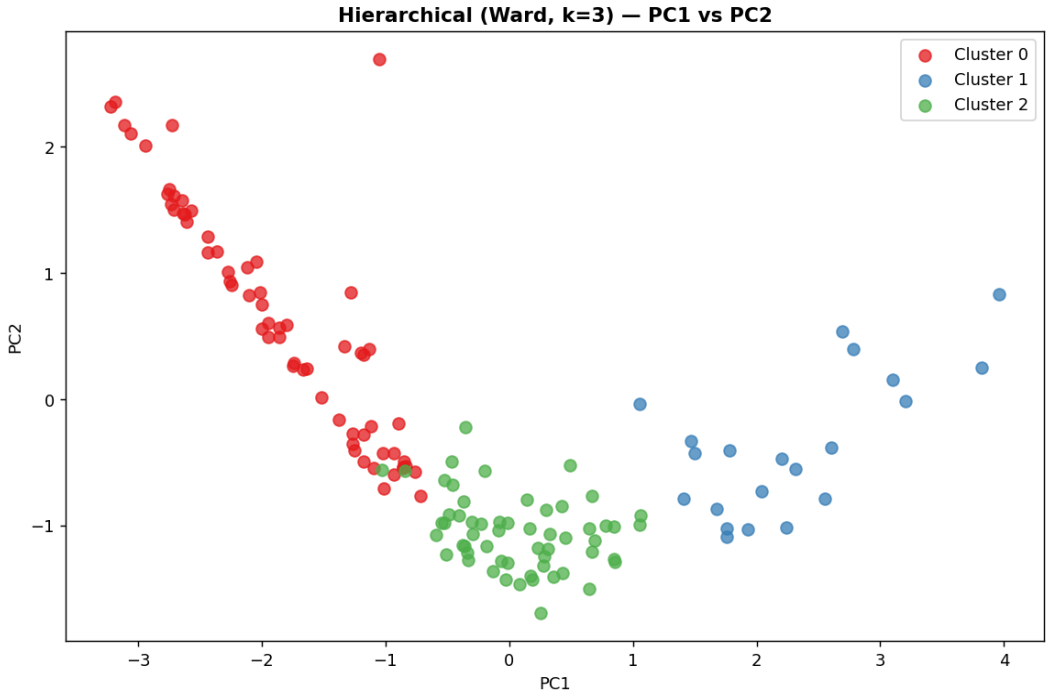


Figure 10. Ward hierarchical clustering (k=3) in PC1–PC2 space — consistent with K-Means.

8. Algorithm Selection and Comparison

Four clustering methods were evaluated. K-Means and Ward hierarchical clustering were selected for the final analysis; GMM and DBSCAN were rejected for the reasons described below.

Method	Silhouette	Decision	Reason
K-Means (k=3)	0.496	Selected	Clear separation; interpretable centroids; silhouette-optimal
Ward HC (k=3)	~0.49	Selected	Confirms K-Means; balanced dendrogram
GMM (k=3)	~0.48	Secondary	Comparable silhouette; soft assignments less interpretable here
DBSCAN	N/A	Rejected	Merges most countries into one cluster; sensitive to eps

Table 3. Comparison of clustering methods and selection rationale.

Although K-Means and Ward HC give slightly different cluster compositions at the margins (a few borderline countries differ in assignment), both agree on the core high-need group. The two methods are therefore used jointly to validate the final recommendation.

9. Final Recommendation

Cluster 2 from the K-Means analysis represents the 44 countries with the most acute development challenges. These countries exhibit a mean child mortality of 87.4 per 1000 live births (15 times higher than the developed cluster), a mean life expectancy of only 59.8 years, a mean GDP per capita of just $1,923, and a mean fertility rate of 5.0 children per woman. These statistics collectively indicate populations with severe, compounding deprivation.

HELP International's $10 million fund should be prioritised for the following 44 countries (K-Means Cluster 2):

Afghanistan, Angola, Benin, Botswana, Burkina Faso, Burundi, Cameroon, Chad, Comoros, Congo (Dem. Rep.), Congo (Rep.), Cote d'Ivoire, Equatorial Guinea, Eritrea, Gabon, Gambia, Ghana, Guinea, Guinea-Bissau, Iraq, Kenya, Kiribati, Lao, Lesotho, Liberia, Madagascar, Malawi, Mali, Mauritania, Mozambique, Namibia, Niger, Pakistan, Rwanda, Senegal, Sierra Leone, Solomon Islands, South Africa, Sudan, Tanzania, Togo, Uganda, Yemen, Zambia.

Additionally, three outlier countries identified in the separate outlier analysis also meet the high-need criteria and should be considered for inclusion: Central African Republic, Haiti, and Nigeria. These were removed as statistical outliers but their development indicators clearly place them within the high-need category.

Within the cluster, the countries with the highest child mortality — Niger, Sierra Leone, Chad, Angola, and Mali — may warrant the most immediate attention. Fund allocation could be weighted by a composite score derived from the cluster indicator means.

10. Conclusion

This analysis applied a rigorous data science pipeline to a 167-country socio-economic dataset. PCA with four components captured 93.4% of the variance and produced uncorrelated inputs for clustering. K-Means (k=3, silhouette = 0.496) and Ward hierarchical clustering converged on the same three-tier structure: developed countries, middle-income developing countries, and a high-need cluster of 44 countries with severely adverse development indicators.

Strengths of the approach:

- Two independent clustering methods produce consistent results, increasing confidence in the partition.

- PCA removes multicollinearity and reduces noise before clustering.
- Outlier countries are not discarded but inspected separately.

Limitations:

- The dataset represents a single point in time; temporal trends are not captured.
- GDP per capita and income are country-level averages; within-country inequality is not reflected.
- The 1.5 × IQR outlier rule is conservative; some genuinely high-need countries were removed and must be added back manually.
- Clustering does not account for aid absorption capacity or governance quality.

Despite these limitations, the clustering analysis provides a transparent, data-driven basis for prioritising humanitarian aid. The 44 recommended countries represent the populations most in need of HELP International's support.